

Christopher Long

SOFTWARE DEVELOPER · CLOUD & SECURITY ENGINEER

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Education

University of Waterloo

September 2021 - August 2026

BCS Computer Science, Honours Co-op

Waterloo, ON, CAN

- Security, Privacy, Cryptography, Operating Systems, Algorithm Design, Application Development, Object-Oriented Programming, Databases

2022 Nortel Institute Scholarship, \$3500 awarded for academic performance

2022 President's Scholarship for Distinction, \$2000 awarded for attaining 95% average or higher

Skills & Tools

Programming Languages

C#, Ruby, Java, Python, Go, C, C++, TypeScript, JavaScript, SQL

Frameworks & Libraries

Ruby on Rails, React, NextJS, Selenium WebDriver, TensorFlow, PyTorch, Tailwind CSS, Bootstrap

Tools & Platforms

Android, AWS, Azure, Docker, JUnit, Firebase, JFrog Artifactory, Linux/Unix, Postman, GitHub Actions

Work Experience

Software Developer — Clio

January 2025 – August 2025

- Engineered OAuth 2.0 authentication flows (Ruby on Rails, ORY Hydra), centralizing identity services across six customer-facing applications.
- Migrated 100% of legacy auth systems to a unified Identity platform, reducing authentication errors and simplifying future MFA rollouts.
- Developed Warden-based SAML and SSO integrations, enabling independent law firms to configure secure, multi-tenant access.
- Collaborated with product, security, and DevOps teams to architect scalable backend services and automate CI/CD pipelines.

VR Developer — Flightline Training Services

May 2024 – December 2024

- Prototyped and implemented interactive VR applications in Unity3D and C#, translating design concepts into production-ready experiences.
- Optimized real-time graphics pipelines for Meta Quest headsets and WebGL, improving frame rates by 20%.
- Integrated photorealistic Blender models and automated asset imports, enhancing immersion and reducing manual cleanup time by 30%.
- Built automated test suites using Unity Test Framework, ensuring stability across multiple device targets.

Cloud Platforms Engineer — Manulife

May 2023 – August 2023

- Automated migration of 10% of API endpoints from Apigee to Microsoft API Management using Node.js, improving request latency by 15%.
- Designed and launched a React dashboard serving 100+ project managers, cutting ticket-wait times by 83% and consolidating approval workflows.
- Configured 10 New Relic synthetic monitors and integrated MoogSoft AIOps, proactively alerting on infrastructure anomalies.
- Supported DevOps reliability efforts by resolving daily PaaS incidents and optimizing Docker- and Azure-based CI/CD pipelines.

Cyber Security Analyst — Manulife

May 2022 – August 2022

- Automated user onboarding workflows in Power Apps and Power Automate, reducing manual steps by 70%.
- Developed a centralized Power Apps portal (SharePoint backend) for 50+ team leads, streamlining employee data management.
- Built an ELT pipeline with Power Automate and Azure Data Factory, feeding live dashboards in Power BI for real-time analytics.
- Conducted security audits on onboarding processes, ensuring compliance with corporate cybersecurity standards.

Projects

Let Me Cook

REPOSITORY LINK

- Designed, led, built and deployed a full-stack recipe book application in Android, with standard CRUD on recipe and account creation, as well as additional AI features using Gemini.
- Heavy compute tasks are delegated to a separate Ktor server, deployed 24/7 on a self-managed Linux VM. Wrote CI/CD workflows to deploy new iterations of the server immediately on new push to repo. API routes and authentication were protected via a JWT dual-token implementation, with rotating server-side secret.

Personal Portfolio

REPOSITORY LINK

- Reactive portfolio page built with Ruby on Rails, Bootstrap, DartSASS-Sprockets.
- Deployed and hosted on Render.

Disable Legacy Auth into Manage and Grow

PRIVATE REPO

- Designed and built flows to replace existing functionalities supported by legacy authentication services.
- Disabled legacy authentication in Clio's 2 largest applications via a killswitch approach, responding to internal and customer feedback.
- Thoroughly tested throughout in development, testing, and staging environments.

User Authentication into Firm Admin

PRIVATE REPO

- Built user authentication methods into Clio's Firm Admin service, according to Senior Software Developer's design spec.
- Implemented using Warden, existing internal Identity SSO service, and SAML authentication available via the internal standards gem.